Beyond Answer Generation: Diagnosing Narrative Reasoning in Transformers

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Abstract

This work investigates whether language models truly understand short stories or rely on surface level patterns. Using the ROC Stories dataset and the Story Cloze Task, four types of incorrect endings are constructed to introduce inconsistencies in temporal order, causal relationships, world state, and emotional tone. Models are evaluated using simple baselines, SBERT, and fine tuned Transformer models including DistilBERT and RoBERTa.

The dataset is found to contain weak surface signals that slightly benefit simple methods. A large gap is observed between performance on synthetic and human written data. Fine tuned models achieve near perfect accuracy on generated endings but perform close to chance on the original Story Cloze task, suggesting reliance on generator specific patterns rather than general story understanding. Temporal reasoning is identified as the most challenging category.

1. Introduction

The Story Cloze Task evaluates whether a model can choose the correct ending to a short story given four sentences of context. While modern language models perform well on this task, it remains unclear whether this reflects true narrative understanding or reliance on surface patterns such as word overlap or emotional tone. This work examines model behavior beyond overall accuracy by analyzing the types of reasoning used and where failures occur. Controlled incorrect endings are constructed that preserve the story's context while introducing specific inconsistencies in temporal order, causal relationships, world state, and emotional tone. Three questions are studied: what surface patterns exist in the dataset, whether models rely on patterns specific to generated endings and generalize to human written data, and which reasoning errors are hardest to detect.

Our hypothesis is: "Models are expected to achieve the highest accuracy on sentiment based inconsistencies, moderate performance on causal inconsistencies, and lower performance on temporal and state inconsistencies, which require integrating information across multiple sentences." The contributions include constructing a dataset of controlled incorrect endings and identifying a large gap between performance on generated and human written data, showing that models often rely on generator specific patterns rather than general story understanding.

2. Background

Several lines of research motivate this work. The Story Cloze Task (Mostafazadeh et al., 2016) was introduced as a benchmark for commonsense reasoning, where models select the correct ending to a short story. While modern transformers achieve high accuracy, it remains unclear whether this reflects true reasoning or reliance on surface patterns. Prior work shows that models often depend on shallow cues. Nie et al. (2020) introduced ANLI, an adversarial dataset designed to remove such shortcuts, leading to significant drops in performance and suggesting that high accuracy may not indicate genuine understanding.

Reasoning over temporal and causal structure remains particularly challenging. Benchmarks such as ExpliCa (2024) and MenatQA (2023) show that models struggle with event ordering and causal relationships. Similarly, Dziri et al. (2023) find that transformers have difficulty with multi-step reasoning and state tracking. Together, these findings suggest that existing benchmarks may overestimate reasoning ability by relying on easy negative examples. This work addresses this limitation by constructing controlled negatives that target specific reasoning types.

3. Exploratory Data Analysis

3.1 Dataset Overview

Dataset Size

Dataset	Count
ROC Stories (full)	52,665 stories
ROC Stories (used, after filtering)	4,841 stories
Story Cloze Validation Set	1,571 stories

Each ROC story has five sentences. Sentences get slightly longer toward the end (7.7 words on average for sentence 1, up to 9.1 for sentence 5). The Story Cloze dataset is balanced: the first ending is correct about 51% of the time.

3.2 Shallow Signal Analysis

Experiment 1: Word Overlap
Correct endings share slightly more words with the story context (0.447 on average) than incorrect endings (0.435). However, the correct ending has higher overlap only 46% of the time. A simple overlap-based model achieves 51.9% accuracy, only slightly above chance.

Experiment 2: Sentiment
A model that selects the ending closest in tone to the story achieves 59.8% accuracy. This indicates that sentiment alignment provides a stronger signal than lexical overlap, but it does not reflect true reasoning.

Experiment 3: Hard Sentiment Cases
In a subset of 390 stories (24.8%) where both endings have similar sentiment, performance drops to 51.3%, close to chance. This suggests that sentiment matching acts as a shortcut rather than a reliable reasoning signal.

3.3 Feature Comparison Table

Feature	Correct Mean	Wrong Mean	Difference
Ending word count	7.59	7.32	+0.26
Lexical overlap	0.447	0.436	+0.012
Jaccard similarity	0.102	0.098	+0.005
New entity count	0.255	0.192	+0.064
Has pronoun	62.8%	57.7%	+5.1%
Has temporal cue	15.0%	9.8%	+5.2%
Has negation	3.6%	9.2%	−5.6%

Correct endings tend to introduce new details and temporal cues, reflecting natural story progression. In contrast, incorrect endings use negation more frequently (9.2% vs 3.6%), suggesting they are often constructed as simple opposites. However, these differences are small and not sufficient for reliable classification.

Summary. The Story Cloze task contains only weak surface signals, indicating that strong performance requires understanding the underlying narrative rather than relying on simple heuristics.

4. Data Generation

Exploratory analysis shows that randomly paired negative endings are often too easy, as they introduce obvious topic mismatches or new entities that can be detected without understanding the story. To address this, controlled negatives are generated using GPT-4o-mini, where each incorrect ending preserves the story's topic and characters while introducing a single targeted reasoning error.

4.1 Creating Wrong Endings with GPT-4o-mini

We used GPT-4o-mini to write one wrong ending per story for each of four categories. Each prompt told the model to: keep the same characters, setting, and topic; write one clear, natural sentence; break only the target rule; not add new characters or places; and not mix up the categories.

Type	**What it breaks**
Temporal	Time order — events happen too early, too late, or out of sequence
Causal	Cause and effect — the outcome does not match what caused it
State	World facts — contradicts something already established in the story
Sentiment	Emotional tone — the ending feels emotionally wrong for the story

Example:

	Text
Context	Tom had a very short temper. One day a guest made him very angry. He punched a hole in the wall of his house. Tom's guest became afraid and left quickly.
Gold ending	Tom sat on his couch filled with regret about his actions.
Temporal	Tom sat on his couch laughing about how he had managed to keep his temper in check during the entire visit.
Causal	Tom felt an overwhelming sense of joy and invited more guests over to celebrate his newfound strength.
State	Tom smiled as he proudly showed off the new mural he had painted over the hole in the wall.
Sentiment	Tom grinned with satisfaction, feeling proud of his strength and the hole he had made in the wall.

4.2 Quality Filtering

We filtered out bad outputs using three rules: (1) length between 4 and 25 words, (2) single sentence only, (3) not identical to the real ending.

Stage	**ROC Stories**	**Cloze Val**
Raw generated rows	20,240	6,284
Failed length filter	211	—
Failed sentence filter	24	—
Final clean rows	19,364	6,284
Stories retained	4,841	1,571

5. Models

5.1 Baseline: Lexical Overlap

This baseline counts the number of shared words between the story context and each candidate ending, selecting the ending with higher overlap. No training or weighting is applied.

5.2 Sentence-BERT (SBERT)

SBERT (all-MiniLM-L6-v2) encodes the context and endings into semantic embeddings. The ending with the highest similarity to the context is selected. No task-specific training is used.

5.3 Pairwise DistilBERT

DistilBERT (distilbert-base-uncased) is fine tuned to compare two candidate endings and select the correct one. Inputs are formatted as [context] [SEP] [ending A] and [context] [SEP] [ending B]. The model is trained for 3 epochs with batch size 16 and learning rate 2e-5.

5.4 Pairwise RoBERTa

RoBERTa (roberta-base) is trained using the same pairwise setup as DistilBERT. Due to pretraining on larger corpora, it is expected to perform better on sentence pair comparison tasks.

5.5 Data Splits

Split	**Source**	**Size**	**Purpose**
Train (80%)	ROC + GPT negatives	3,872 stories / 15,488 pairs	Train DistilBERT and RoBERTa
Monitor (20%)	ROC + GPT negatives	969 stories / 3,876 pairs	Early stopping
Val B	Cloze Val (human endings)	1,571 stories	Real task accuracy
Val C	Cloze Val + GPT negatives	1,571 × 4 types	Per-type fake detection

6. Results

6.1 Val B: Real Story Cloze Accuracy

Model	**Val B Accuracy**	**p-value**	**Significant?**
Lexical Overlap (Baseline)	51.7%	0.095	No
SBERT	59.4%	< 0.001	Yes
DistilBERT	50.6%	0.325	No
RoBERTa	57.0%	< 0.001	Yes

A one sided binomial test is used to evaluate performance against chance (50%), with $n = 1,571$. SBERT achieves the highest accuracy on the real task. RoBERTa is the only fine tuned model that clearly exceeds chance, while DistilBERT performs near chance despite being trained on the task.

6.2 Val C: Fake Ending Detection by Type

Model	Temporal	Causal	State	Sentiment	Overall
Lexical Overlap	6.6%	9.5%	11.3%	13.4%	10.2%
SBERT	23.8%	41.1%	44.3%	54.2%	40.9%
DistilBERT	99.3%	99.8%	98.6%	98.8%	99.1%
RoBERTa	99.7%	99.9%	99.5%	99.7%	99.7%

6.3 Training Curves (Fine-tuned Models)

DistilBERT:

Epoch	Train Loss	Val Loss	Val Acc
1	0.1230	0.0329	99.15%
2	0.0267	0.0244	99.45%
3	0.0109	0.0289	99.41%

RoBERTa:

Epoch	Train Loss	Val Loss	Val Acc
1	0.0768	0.0374	99.33%
2	0.0156	0.0300	99.47%
3	0.0039	0.0332	99.55%

Both models learn very quickly and hit near-perfect accuracy on the training set. This suggests they found an easy signal rather than learning to reason about stories.

7. Analysis

7.1 The Val B vs Val C Gap — Main Finding

Model	Val B	Val C	Gap (C − B)
Lexical Overlap	51.7%	10.2%	−41.5%
SBERT	59.4%	40.9%	−18.5%
DistilBERT	50.6%	99.1%	+48.5%
RoBERTa	57.0%	99.7%	+42.7%

DistilBERT and RoBERTa score over 99% at detecting GPT-generated fake endings, but only 50–57% on the real task. This means they learned to recognize how GPT-4o-mini writes, not how stories work. When tested on human-written alternatives, they fall apart.RoBERTa's smaller gap (42.7% vs 48.5%) suggests it has slightly more general understanding built in from pretraining.

7.2 Which Reasoning Type is Hardest?

For the two models with no training (Lexical Overlap and SBERT), time-order errors were the hardest to catch and sentiment errors were the easiest:

Model	Temporal Error	Causal Error	State Error	Sentiment Error
Lexical Overlap	93.4%	90.5%	88.7%	86.6%
SBERT	76.2%	58.9%	55.7%	45.8%

Time-order errors are hard because you need to track when things happen across multiple sentences. Sentiment errors are easier because emotional words appear directly in the text.

7.3 Error Examples

Here are examples of wrong endings that fooled the Lexical Overlap model:

Temporal (events in the wrong order)
Context: “Laverne follows a brownie recipe closely. She tests one brownie.”
Real ending: “The brownies are so delicious Laverne eats two of them.”
Fake ending: “Laverne tests one brownie after she has already eaten all of them.”

Causal (outcome does not follow from the story)
Context: “John and Billy won a beer pong contest. The next level sent them to Vegas.”
Real ending: “In Vegas, they competed against eighty contestants.”
Fake ending: “In Vegas, they decided to quit beer pong and take up professional knitting.”

State (contradicts an established fact)
Context: “Ron finishes landscaping the mayor’s yard early and is ecstatic.”
Real ending: “His boss commends him for a job well done.”
Fake ending: “His boss is unhappy and fires Ron on the spot.”

Sentiment (emotional tone clashes with the story):

Context: “Luke was playing hockey at school. The game was tied and almost over. Then Luke made the winning shot! Everybody cheered!”

Real ending: “Luke was so proud of himself!”

Fake ending: “Luke felt utterly defeated and wished he had never played.”

8. Discussion

The key problem we found: when the training data and the test fake endings both come from the same AI (GPT-4o-mini), the model just learns GPT's writing patterns. It does not actually learn story logic. The gap between Val B and Val C scores shows this clearly.RoBERTa does a bit better on the real task (57%), but the large gap shows it still relies on style matching. The most honest result comes from SBERT: 59.4% on the real task without any fine-tuning. This shows that good sentence representations can help even without task-specific training.

9. Conclusion

A framework is developed to evaluate how models handle four types of story errors. Fine tuned models achieve near perfect accuracy on AI generated negative endings but perform only slightly above chance on human written ones, suggesting that they learn generator specific patterns rather than underlying narrative reasoning. SBERT and RoBERTa are the only models that clearly exceed chance on the real task, and temporal reasoning emerges as the most challenging category across all models. Future work can explore using human written negative endings and evaluating models trained on one generator against outputs from another, as well as generating negatives with different language models to better separate style from reasoning.

Authors' Contributions

Nitya Sree Cheera: conducted the shallow signal EDA (Section 3.2), including the word overlap, sentiment, and hard sentiment subset experiments. Nitya implemented Models 1 (Lexical Overlap), 4 (SBERT), 5 (DistilBERT), and 6 (RoBERTa), and led the GPT-4o-mini negative generation pipeline.

Aurelia Yang: conducted the feature comparison EDA (Section 3.3), analyzing structural and linguistic differences between correct and wrong endings. Aurelia also contributed to model development through experimental design ideas and feedback, created the full slide deck for the final presentation, and produced additional visualizations to highlight the main findings.

Both team members contributed to the experimental design, analysis, and writing of this report.

References

1.Mostafazadeh, N., Chambers, N., He, X., Parikh, D., Batra, D., Vanderwende, L., Kohli, P., & Allen, J. (2016). A corpus and evaluation framework for deeper understanding of commonsense stories. NAACL-HLT 2016. https://arxiv.org/abs/1604.01696

2. Nie, Y., Williams, A., Dinan, E., Bansal, M., Weston, J., & Kiela, D. (2020). Adversarial NLI: A new benchmark for natural language understanding. ACL 2020. https://arxiv.org/abs/1910.14599

3. Dziri, N., Lu, X., Sclar, M., Li, X. L., Jiang, L., Lin, B. Y., West, P., Bhagavatula, C., Bras, R. L., Hwang, J. D., Sanyal, S., Welleck, S., Etzioni, O., Harchaoui, Z., Sap, M., & Choi, Y. (2023). Faith and fate: Limits of transformers on compositionality. NeurIPS 2023. https://arxiv.org/abs/2305.18654

4. Wei, C., Zhang, Y., Wan, X., & Li, P. (2023). MenatQA: A new dataset for testing the temporal comprehension and reasoning abilities of large language models. EMNLP Findings 2023. https://arxiv.org/abs/2310.05157

5. Chen, Z., & Gao, X. (2024). ExpliCa: Evaluating explicit causal reasoning in large language models. https://arxiv.org/abs/2402.09752

Addendum: Zero-Shot GPT-4o Evaluation on Val B

Updated Val B: Real Story Cloze Accuracy

Model	Val B Acc	p-value	Significant?
Lexical Overlap	51.7%	0.095	No
SBERT	59.4%	< 0.001	Yes
DistilBERT	50.6%	0.325	No
RoBERTa	57.0%	< 0.001	Yes
GPT-4o (zero-shot)	**99.6%**	**< 0.001**	**Yes**

Updated Val B vs Val C Gap

Model	Val B	Val C (Overall)	Gap (C · B)
Lexical Overlap	51.7%	10.2%	·41.5%
SBERT	59.4%	40.9%	·18.5%
DistilBERT	50.6%	99.1%	+48.5%
RoBERTa	57.0%	99.7%	+42.7%
GPT-4o (zero-shot)	**99.6%**	**N/A**	**N/A**

Discussion

Appendix A: Additional Baselines

A.1 Model 2 — TF-IDF Cosine Similarity

This model turns the story context and each ending into weighted word vectors using TF-IDF. It picks the ending whose vector is closest to the context. No training is needed.

Results: - Val B: 51.7% ($p = 0.095$, not significant) - Val C: Temporal 31.0%, Causal 48.0%, State 47.4%, Sentiment 55.3%, Overall 45.4% - Only sentiment is statistically significant ($p < 0.001$)

A.2 Model 3 — TF-IDF + Logistic Regression

A classifier trained to tell correct endings from wrong ones using TF-IDF features.

Results: - Train accuracy: 90.6% (30,976 pairs) - Monitor accuracy: 80.4% (7,752 pairs) - Val B: 59.3% (significant, $p < 0.001$) - Val C: Temporal 98.3%, Causal 97.7%, State 96.1%, Sentiment 97.3%, Overall 97.4%This model shows the same Val B vs Val C gap as the transformer models. Even a simple classifier can learn GPT's writing style.

Appendix B: Full Results Table

Model	**Val B**	**C-Temp**	**C-Causal**	**C-State**	**C-Sent**	**C-Overall**
Lexical Overlap	51.7%	6.6%	9.5%	11.3%	13.4%	10.2%
TF-IDF Cosine	51.7%	31.0%	48.0%	47.4%	55.3%	45.4%
TF-IDF + LR	59.3%	98.3%	97.7%	96.1%	97.3%	97.4%
SBERT	59.4%	23.8%	41.1%	44.3%	54.2%	40.9%
DistilBERT	50.6%	99.3%	99.8%	98.6%	98.8%	99.1%
RoBERTa	57.0%	99.7%	99.9%	99.5%	99.7%	99.7%